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Course : T.Y.B.Sc.I.T.
Div : A
Roll No. : 13
Subject : Advanced Mobile Programming
Year : 2023 - 2024



UNIVERSITY OF MUMBAI
SHETH T.J. EDUCATION SOCIETY'S,
SHETH N.K.T.T. COLLEGE OF COMMERCE &
SHETH J.T.T. COLLEGE OF ARTS, THANE (W)

CERTIFICATE

This is to certify that Mr./~~Miss~~. Aayush Rajesh More
of B.Sc. (IT) Semester VI has completed the practical work in the subject of
“Advanced Mobile Programming” during the academic year 2023-24 under the
guidance of “Mrs. Sneha Gupta” being the partial requirement for the fulfillment
of the Curriculum of Degree of Bachelor of Science in Information Technology,
University of Mumbai.

Place : THANE

Date : _____

Internal Examiner

External Examiner

Co-ordinator

INDEX

Sr.No	Practicals	Date	Sign
1	Introduction to Android, Introduction to Android Studio IDE, Application Fundamentals: Creating a Project, Android Components, Activities, Services, Content Providers, Broadcast Receivers, Interface overview, Creating Android Virtual device, USB debugging mode, Android Application Overview. Simple “Hello World” program.		
2	Programming Resources Android Resources: (Color, Theme, String, Drawable, Dimension, Image),		
3	Programming Activities and fragments Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.		
4	Programs related to different Layouts Coordinate, Linear, Relative, Table, Absolute, Frame, List View, Grid View.		
5	Programming UI elements AppBar, Fragments, UI Components		
6	Programming menus, dialog, dialog fragments		
7	Programs on Intents, Events, Listeners and Adapters The Android Intent Class, Using Events and Event Listeners		
8	Database Programming with SQLite		

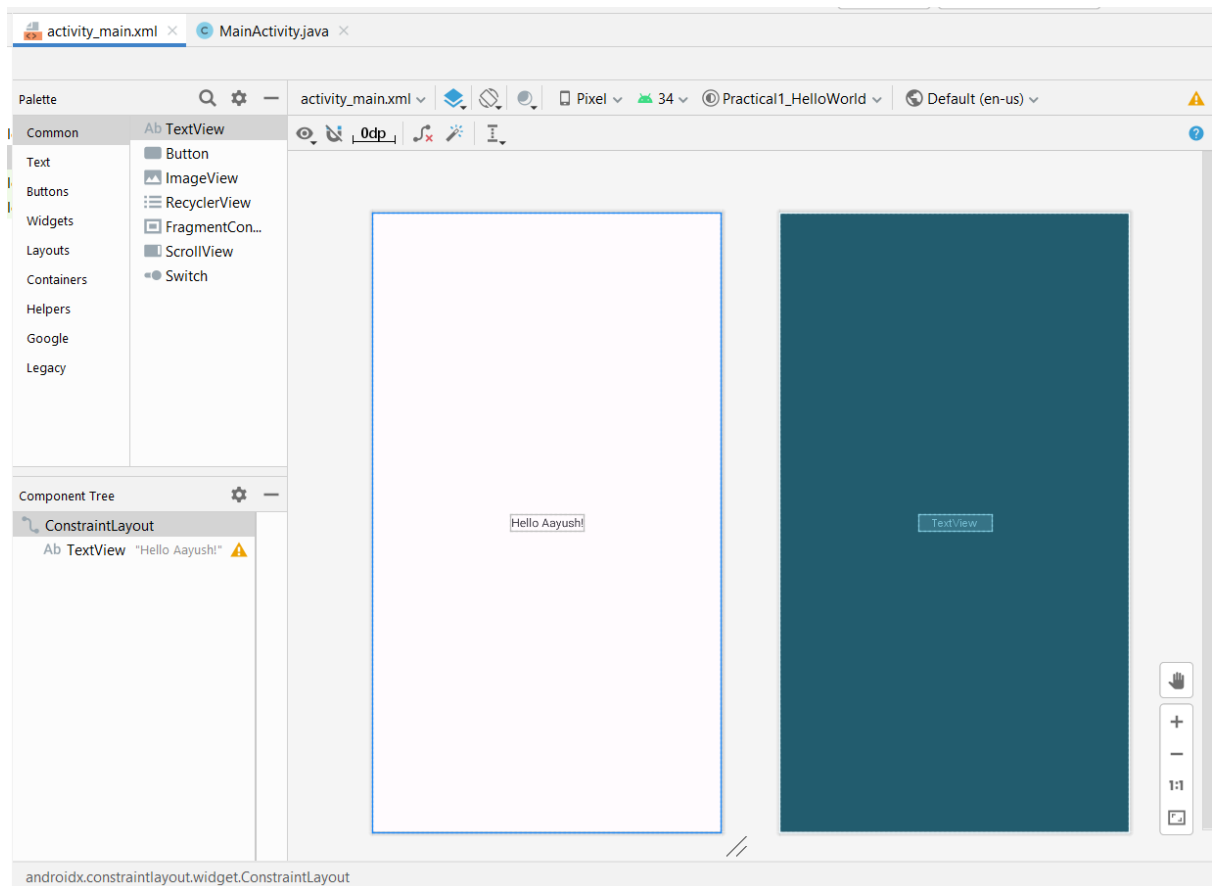
1. Simple “Hello World” program.

Activity_Main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Aayush!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```



MainActivity.java

```
package com.example.practical1_helloworld;

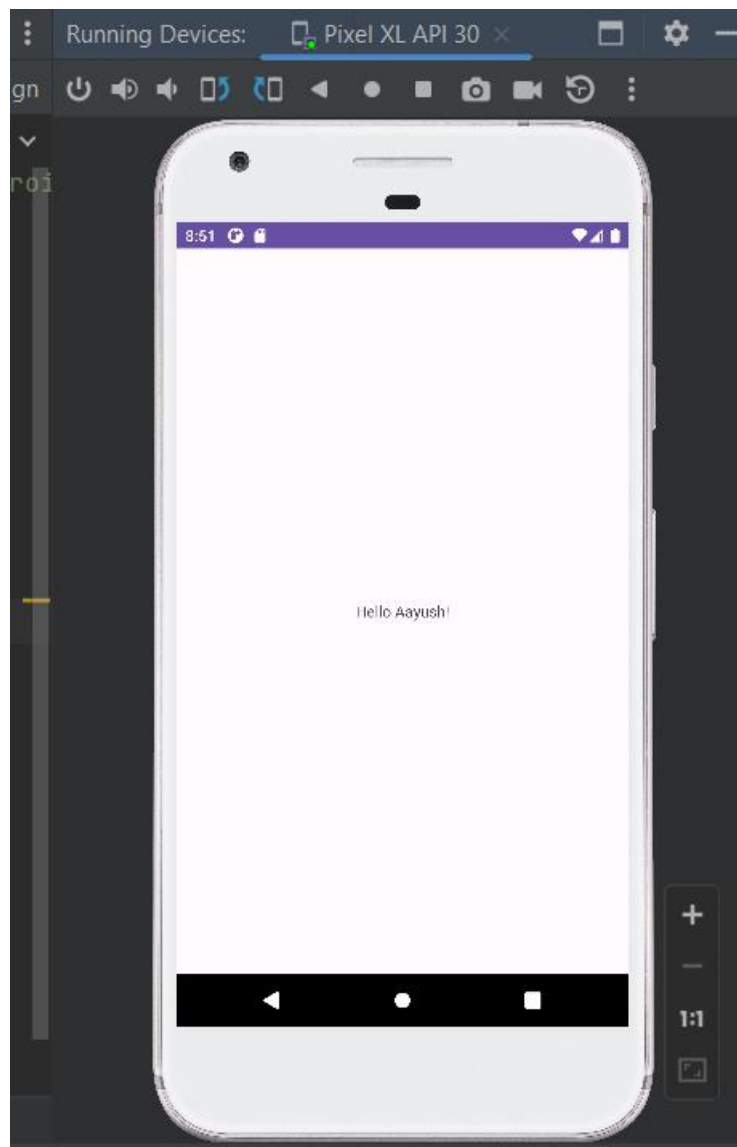
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output :



2. Programming Android Resources: (Color, Theme, String, Drawable, Dimension, Image)

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/xyz"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/Aayush"
        android:textSize="50dp"
        android:textStyle="bold"
        android:textColor="@color/Aayush_Green"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Main_Activity.java

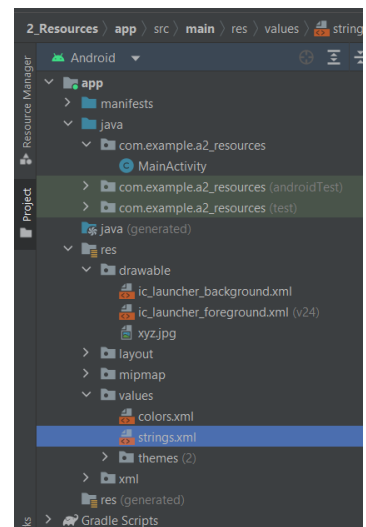
```
package com.example.a2_resources;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```



Strings.xml

```
<resources>
    <string name="app_name">2_Resources</string>
    <string name="Aayush">
        <item>Hello</item>
        <item>Aayush</item>
    </string>
</resources>
```

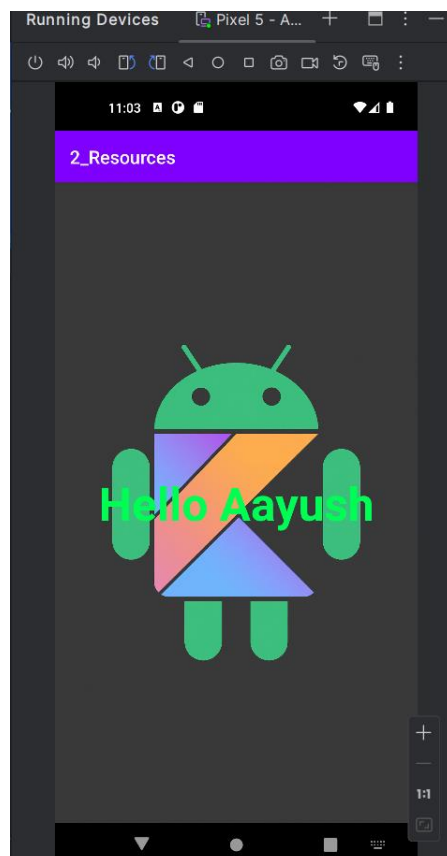
Colours.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="Aayush_Blue">#03A9F4</color>
    <color name="Aayush_Green">#00FF55</color>
</resources>
```

Themes.xml

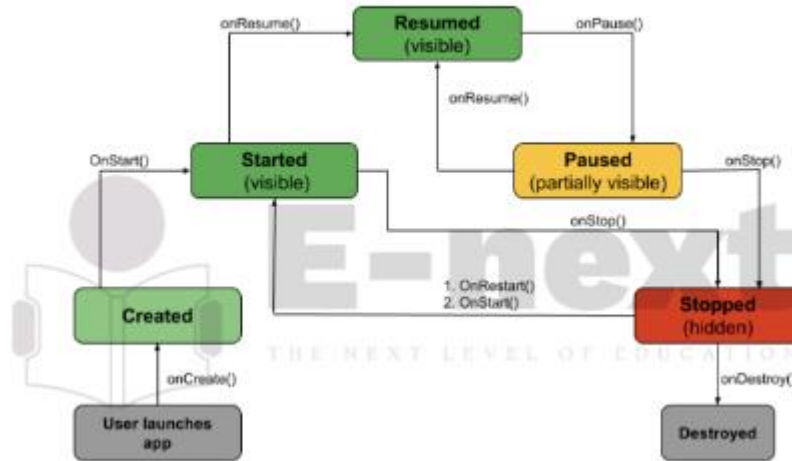
```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.2_Resources"
parent="Theme.AppCompat.DayNight.DarkActionBar">
    <!-- Customize your light theme here. -->
    <item name="colorPrimary">#8000FF</item>
    </style>
    <style name="Theme.2_Resources" parent="Base.Theme.2_Resources" />
</resources>
```

Output:



3. Programming Activities and fragments: Activity Life Cycle, Activity methods, Multiple Activities

Activity Lifecycle:



- **onCreate():** Called by the OS when the activity is first created. This is where you initialize any UI elements or data objects. You also have the `savedInstanceState` of the activity that contains its previously saved state, and you can use it to recreate that state.
- **onStart():** Just before presenting the user with an activity, this method is called. It's always followed by `onResume()`. In here, you generally should start UI animations, audio based content or anything else that requires the activity's contents to be on screen.
- **onResume():** As an activity enters the foreground, this method is called. Here you have a good place to restart animations, update UI elements, restart camera previews, resume audio/video playback or initialize any components that you release during `onPause()`.
- **onPause():** This method is called before sliding into the background. Here you should stop any visuals or audio associated with the activity such as UI animations, music playback or the camera. This method is followed by `onResume()` if the activity returns to the foreground or by `onStop()` if it becomes hidden.
- **onStop():** This method is called right after `onPause()`, when the activity is no longer visible to the user, and it's a good place to save data that you want to commit to the disk. It's followed by either `onRestart()`, if this activity is coming back to the foreground, or `onDestroy()` if it's being released from memory.
- **onRestart():** Called after stopping an activity, but just before starting it again. It's always followed by `onStart()`.
- **onDestroy():** This is the final callback you'll receive from the OS before the activity is destroyed. You can trigger an activity's destruction by calling `finish()`, or it can be triggered by the system when the system needs to recoup memory. If your activity includes any background threads or other long-running resources, destruction could lead to a memory leak if they're not released, so you need to remember to stop these processes here as well.

Activity_Main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

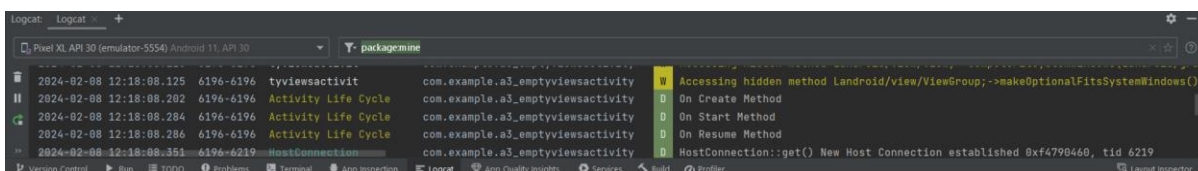
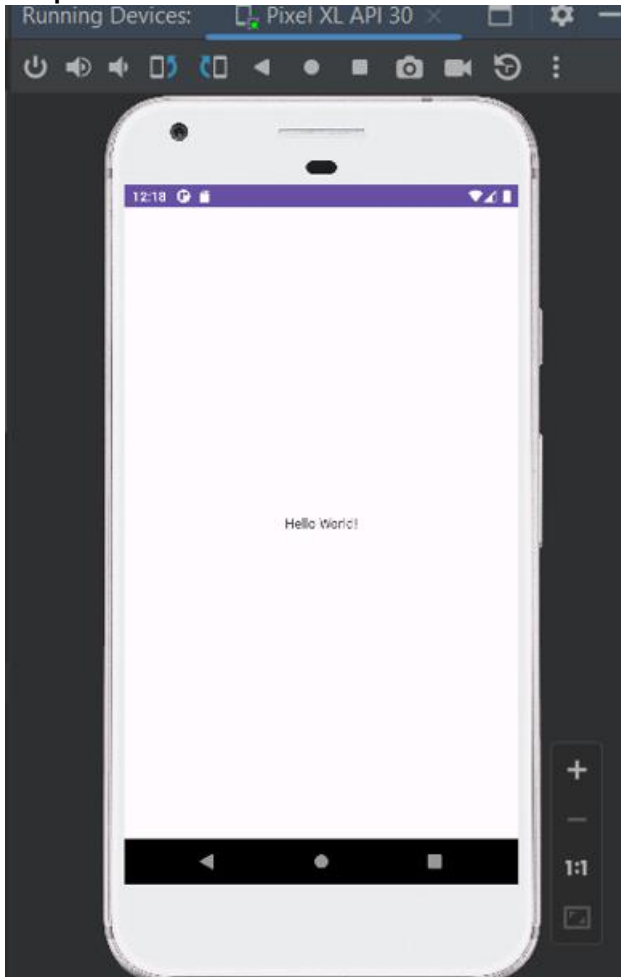
MainActivity.java

```
package com.example.a3_emptyviewsactivity;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;
public class MainActivity extends AppCompatActivity {
    String tag = "Activity Life Cycle ";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d(tag,"On Create Method");
    }
    @Override
    protected void onStart() {
        super.onStart();
        Log.d(tag,"On Start Method");
    }
    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d(tag,"On Restart Method");
    }

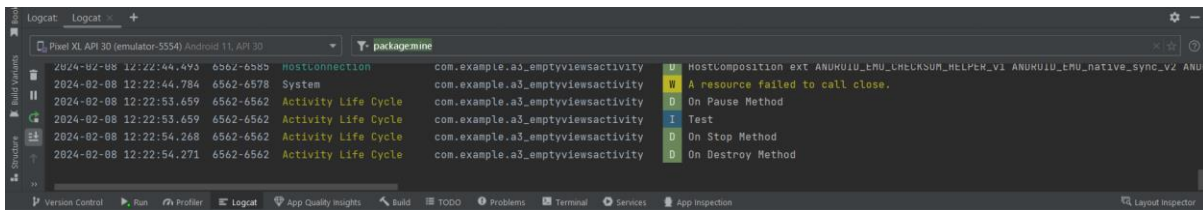
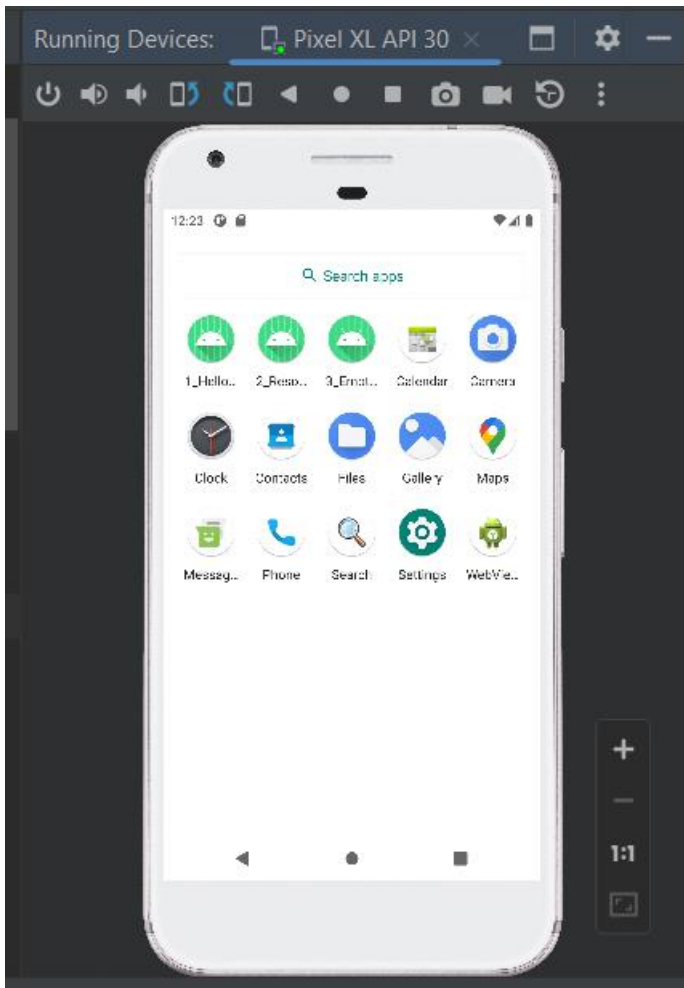
    @Override
    protected void onPause() {
        super.onPause();
        Log.d(tag,"On Pause Method");
        Log.i(tag,"Test");
    }
}
```

```
}
@Override
protected void onResume() {
    super.onResume();
    Log.d(tag, "On Resume Method");
}
@Override
protected void onStop() {
    super.onStop();
    Log.d(tag, "On Stop Method");
}
@Override
protected void onDestroy() {
    super.onDestroy();
    Log.d(tag, "On Destroy Method");
}
}
```

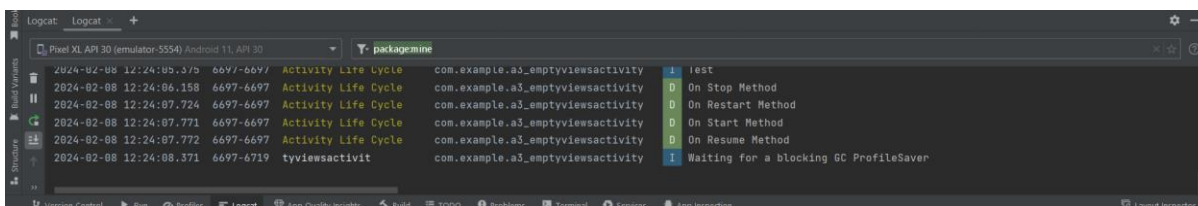
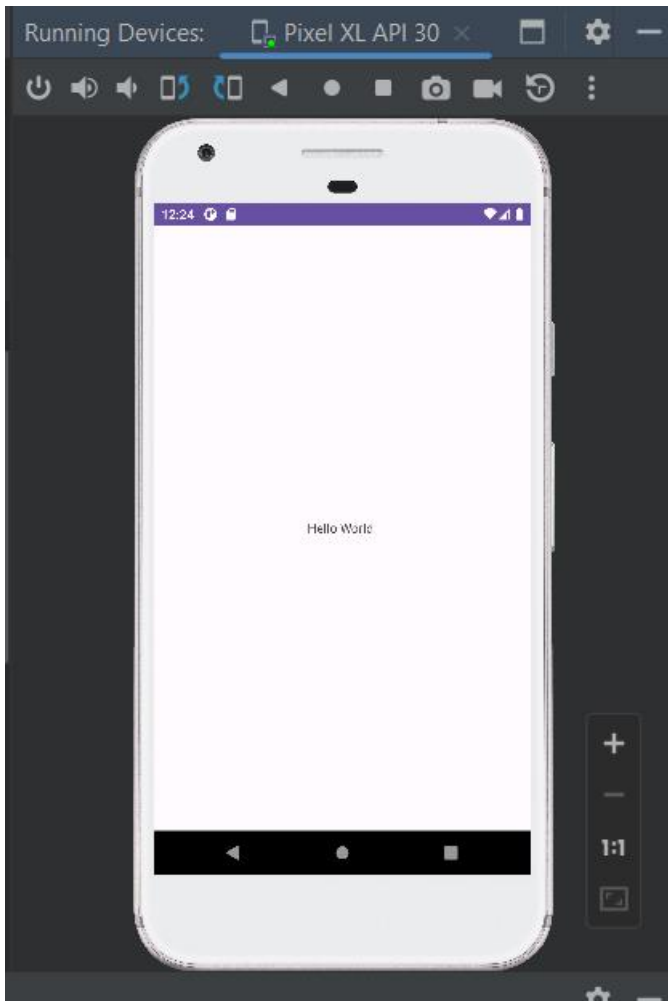
Output:



Advanced Mobile Programming



Advanced Mobile Programming



4. Programs related to different layouts

4a)_ Linear_Layout

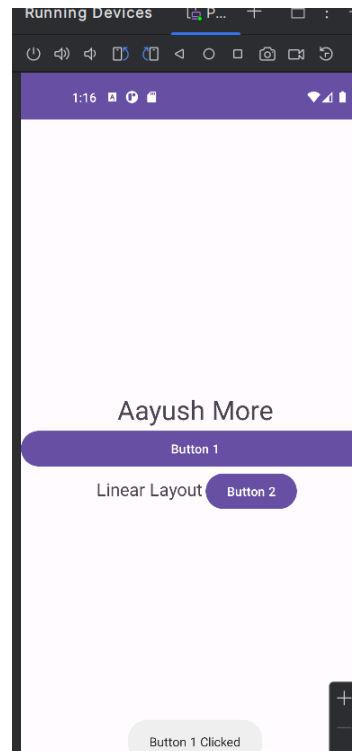
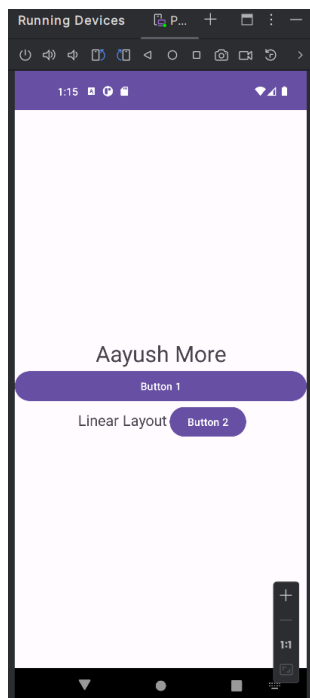
Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        >
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="30dp"
        android:text="Aayush More" />
    <Button
        android:id="@+id/btn1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Button 1" />
    </LinearLayout>
    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        >
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textAlignment="center"
        android:padding="10px"
        android:textSize="20dp"
        android:text="Linear Layout" />
    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2" />
    </LinearLayout>
</LinearLayout>
```

Main_Activity.java :

```
package com.example.tablelayout;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    private Button btnClick;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnClick = (Button) findViewById(R.id.btn1);
        btnClick.setOnClickListener(new View.OnClickListener() {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "Button 1 CLicked", Toast.LENGTH_SHORT).show();
            }
        });
        btnClick = (Button) findViewById(R.id.btn2);
        btnClick.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "Button 2 CLicked", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Output :



4b)_ RelativeLayout

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
>
    <LinearLayout
        android:orientation="vertical"
        android:layout_width="fill_parent"
        android:layout_height="fill_parent"
        android:layout_alignParentStart="true"
        android:gravity="center"
    >
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textAlignment="center"
            android:textSize="30dp"
            android:text="Aayush More" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textAlignment="center"
            android:textSize="30dp"
            android:text="Relative Layout" />

        <Button
            android:id="@+id/btn1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 1" />

        <Button
            android:id="@+id/btn2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 2" />

    </LinearLayout>

</RelativeLayout>
```

Advanced Mobile Programming

Main_Activity.java :

```
package com.example.tablelayout;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

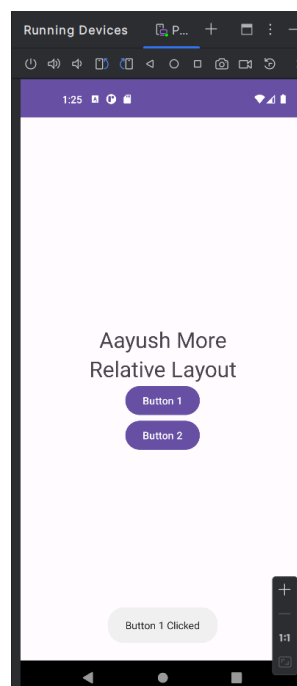
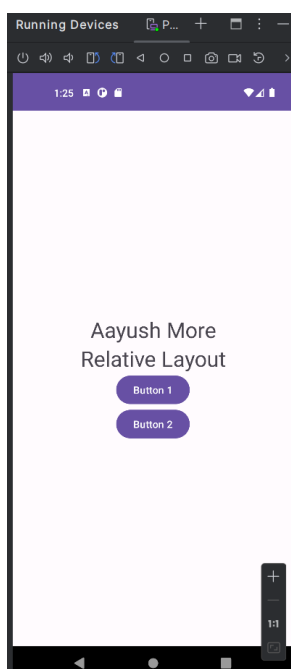
public class MainActivity extends AppCompatActivity {
    private Button btnClick;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnClick = (Button) findViewById(R.id.btn1);
        btnClick.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "Button 1 Clicked", Toast.LENGTH_SHORT).show();
            }
        });

        btnClick = (Button) findViewById(R.id.btn2);
        btnClick.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "Button 2 Clicked", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Output :



4c)_ Table Layout

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="30dp"
        android:text="Table Layout" />

    <TableLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">

        <TableRow
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">

            <Button
                android:id="@+id/btn1"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="1" />

            <Button
                android:id="@+id/btn2"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="2" />

            <Button
                android:id="@+id/btn3"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="3" />
        </TableRow>

        <TableRow
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">
```

```
<Button
    android:id="@+id/btn4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="4" />

<Button
    android:id="@+id/btn5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="5" />

<Button
    android:id="@+id/btn6"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="6" />
</TableRow>

<TableRow
    android:layout_width="wrap_content"
    android:layout_height="wrap_content">

    <Button
        android:id="@+id/btn7"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="7" />

    <Button
        android:id="@+id/btn8"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="8" />

    <Button
        android:id="@+id/btn9"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="9" />
</TableRow>

</TableLayout>
</LinearLayout>
```

Main_Activity.java :

```
package com.example.tablelayout;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.widget.Button;
```

```
import android.widget.Toast;
```

```
public class MainActivity extends AppCompatActivity {  
    private Button btnClick;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);
```

```
        btnClick = (Button) findViewById(R.id.btn1);  
        btnClick.setOnClickListener(new View.OnClickListener()  
        {  
            public void onClick(View view)  
            {  
                Toast.makeText(getApplicationContext(), "1", Toast.LENGTH_SHORT).show();  
            }  
        });
```

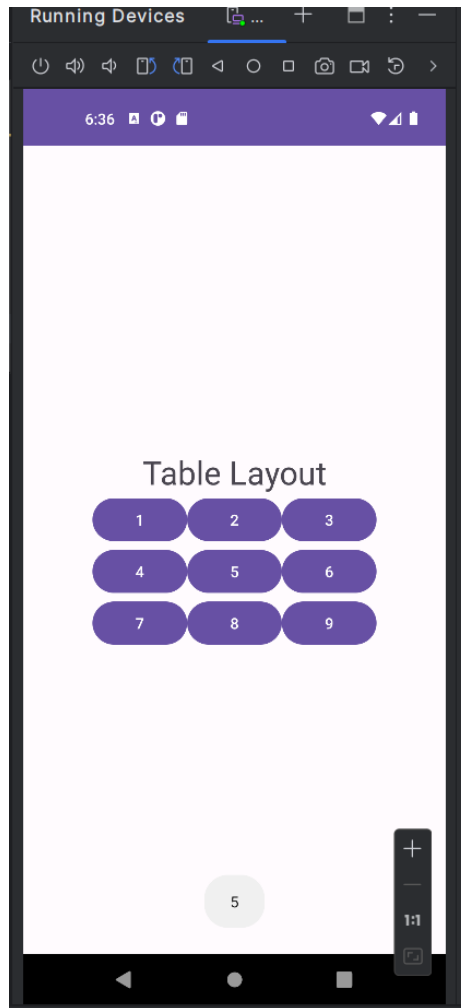
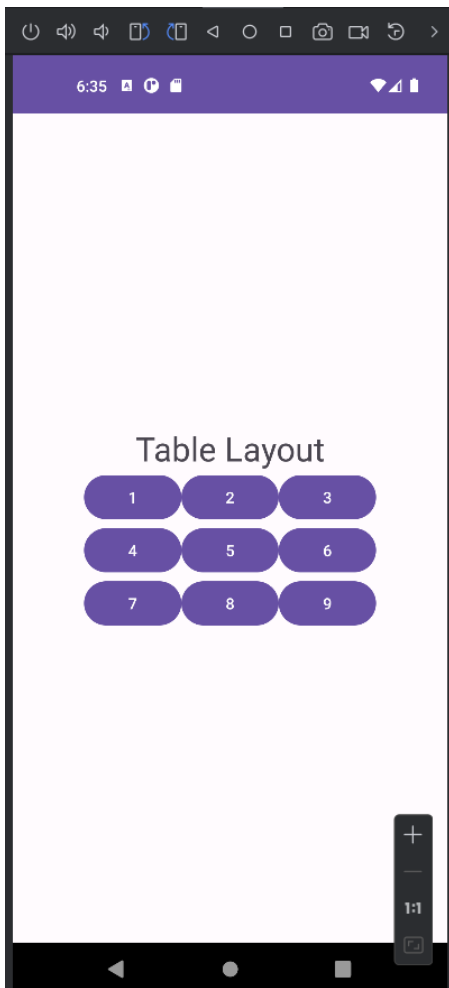
```
        btnClick = (Button) findViewById(R.id.btn2);  
        btnClick.setOnClickListener(new View.OnClickListener()  
        {  
            public void onClick(View view)  
            {  
                Toast.makeText(getApplicationContext(), "2", Toast.LENGTH_SHORT).show();  
            }  
        });
```

```
        btnClick = (Button) findViewById(R.id.btn3);  
        btnClick.setOnClickListener(new View.OnClickListener()  
        {  
            public void onClick(View view)  
            {  
                Toast.makeText(getApplicationContext(), "3", Toast.LENGTH_SHORT).show();  
            }  
        });
```

```
        btnClick = (Button) findViewById(R.id.btn4);  
        btnClick.setOnClickListener(new View.OnClickListener()  
        {  
            public void onClick(View view)  
            {  
                Toast.makeText(getApplicationContext(), "4", Toast.LENGTH_SHORT).show();
```

```
    }  
});  
  
btnClick = (Button) findViewById(R.id.btn5);  
btnClick.setOnClickListener(new View.OnClickListener()  
{  
    public void onClick(View view)  
    {  
        Toast.makeText(getApplicationContext(), "5", Toast.LENGTH_SHORT).show();  
    }  
});  
  
btnClick = (Button) findViewById(R.id.btn6);  
btnClick.setOnClickListener(new View.OnClickListener()  
{  
    public void onClick(View view)  
    {  
        Toast.makeText(getApplicationContext(), "6", Toast.LENGTH_SHORT).show();  
    }  
});  
  
btnClick = (Button) findViewById(R.id.btn7);  
btnClick.setOnClickListener(new View.OnClickListener()  
{  
    public void onClick(View view)  
    {  
        Toast.makeText(getApplicationContext(), "7", Toast.LENGTH_SHORT).show();  
    }  
});  
  
btnClick = (Button) findViewById(R.id.btn8);  
btnClick.setOnClickListener(new View.OnClickListener()  
{  
    public void onClick(View view)  
    {  
        Toast.makeText(getApplicationContext(), "8", Toast.LENGTH_SHORT).show();  
    }  
});  
  
btnClick = (Button) findViewById(R.id.btn9);  
btnClick.setOnClickListener(new View.OnClickListener()  
{  
    public void onClick(View view)  
    {  
        Toast.makeText(getApplicationContext(), "9", Toast.LENGTH_SHORT).show();  
    }  
});  
}  
}
```

Output :



4d)_ FrameLayout

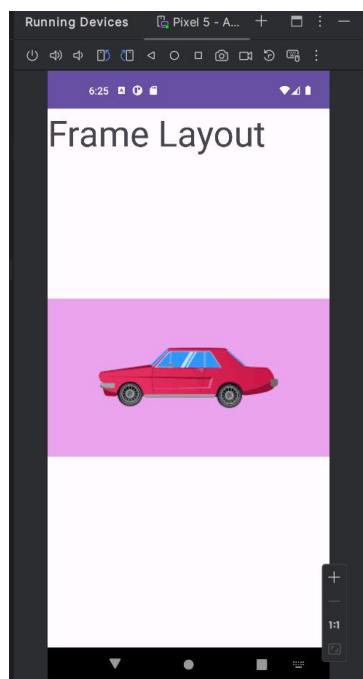
Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" >
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="50sp"
        android:textAlignment="center"
        android:text="Frame Layout" />
    <ImageView
        android:layout_height="match_parent"
        android:layout_width="match_parent"
        android:src="@drawable/x" > </ImageView>
</FrameLayout>
```

Main_Activity.java :

```
package com.example.framelayout;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output :



4e)_ Grid Layout

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:columnCount="3"
    android:rowCount="3"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="1" />

    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="2" />

    <Button
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="3" />

    <Button
        android:id="@+id/btn4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="4" />

    <Button
        android:id="@+id/btn5"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="5" />

    <Button
        android:id="@+id/btn6"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="6" />

    <Button
        android:id="@+id/btn7"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="7" />
```

```
<Button
    android:id="@+id/btn8"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="8" />
```

```
<Button
    android:id="@+id/btn9"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="9" />
```

```
</GridLayout>
```

Main_Activity.java :

```
package com.example.tablelayout;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    private Button btnClick;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnClick = findViewById(R.id.btn1);
        btnClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                showToast("Button 1 clicked");
            }
        });

        btnClick = findViewById(R.id.btn2);
        btnClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                showToast("Button 2 clicked");
            }
        });

        btnClick = findViewById(R.id.btn3);
        btnClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                showToast("Button 3 clicked");
            }
        });
    }
}
```



```
});

btnClick = findViewById(R.id.btn4);
btnClick.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        showToast("Button 4 clicked");
    }
});

btnClick = findViewById(R.id.btn5);
btnClick.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        showToast("Button 5 clicked");
    }
});

btnClick = findViewById(R.id.btn6);
btnClick.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        showToast("Button 6 clicked");
    }
});

btnClick = findViewById(R.id.btn7);
btnClick.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        showToast("Button 7 clicked");
    }
});

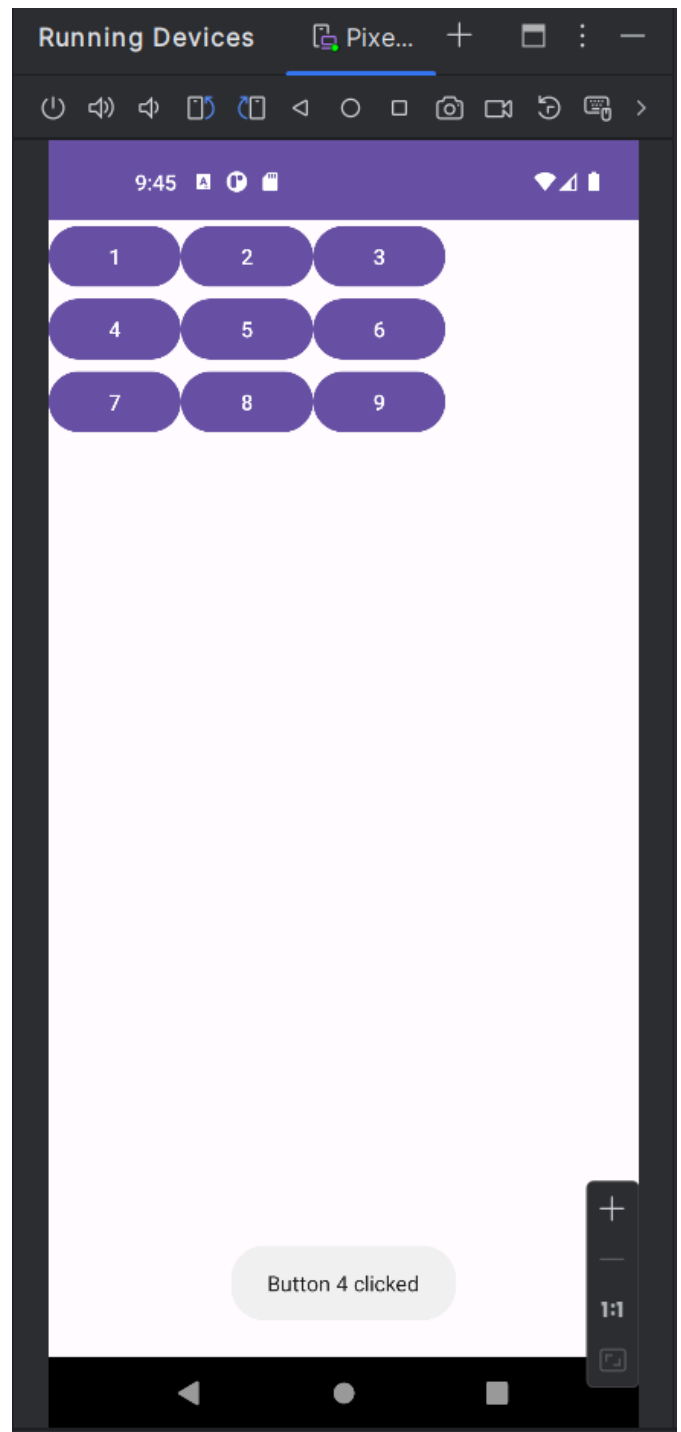
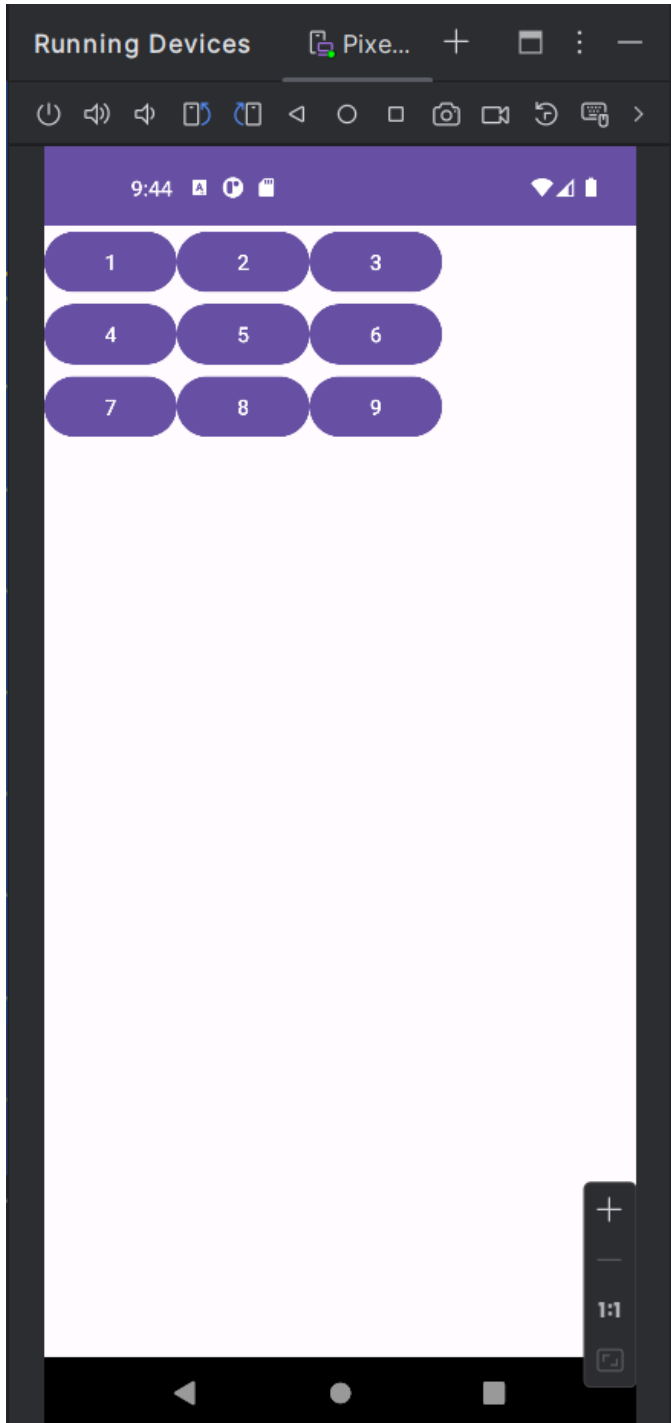
btnClick = findViewById(R.id.btn8);
btnClick.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        showToast("Button 8 clicked");
    }
});

btnClick = findViewById(R.id.btn9);
btnClick.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        showToast("Button 9 clicked");
    }
});
}
```

Advanced Mobile Programming

```
private void showToast(String message) {  
    Toast.makeText(getApplicationContext(), message, Toast.LENGTH_SHORT).show();  
}  
}
```

Output :



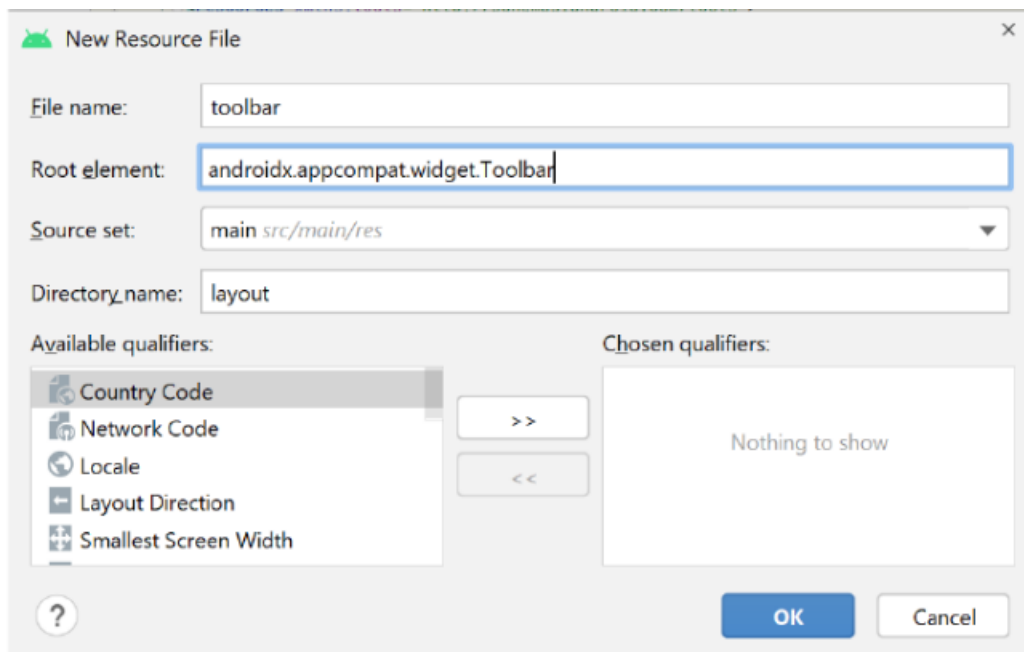
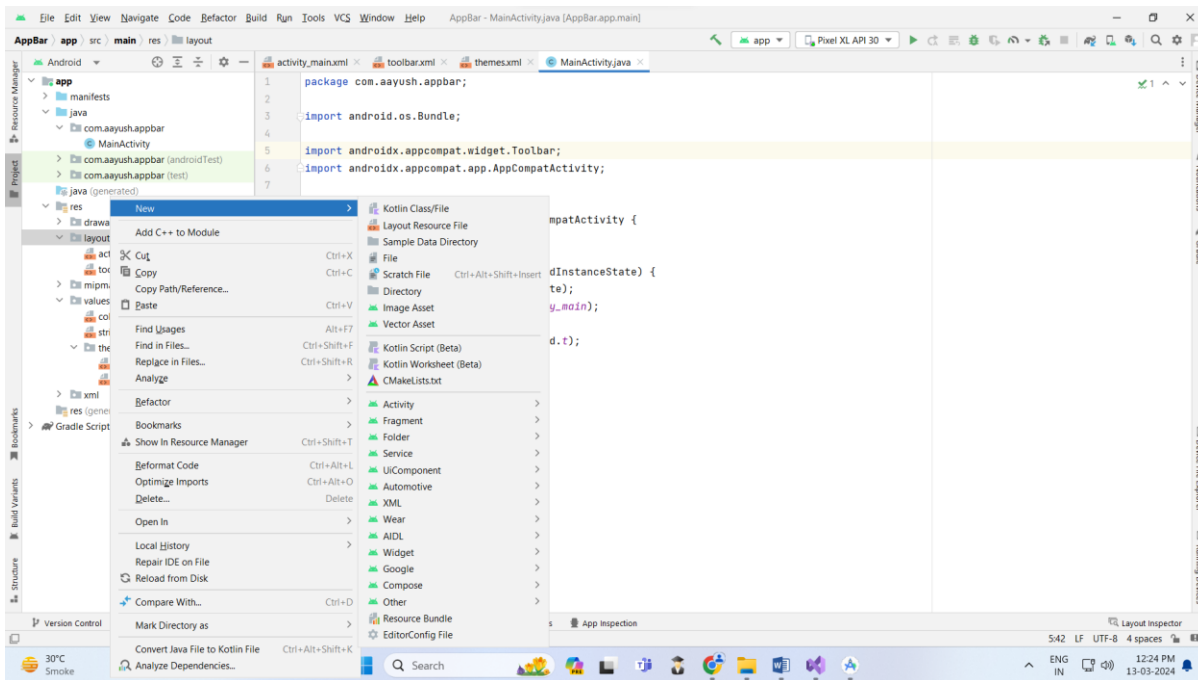
PRACTICAL 5

Programming UI elements AppBar, Fragments, UI Components

5a. Demonstration of Application Bar

ProjectName -> App -> Src -> Main -> Res -> values-> styles.xml The default line is: `<style name="Theme.Practical5" parent="Theme.MaterialComponents.DayNight.DarkActionBar">` Change to:

`<style name="Theme.Practical5" parent="Theme.MaterialComponents.Light.NoActionBar">`



Themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.AppBar"
parent="Theme.MaterialComponents.Light.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.AppBar" parent="Base.Theme.AppBar" />
</resources>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <include android:id="@+id/t" layout="@layout/toolbar" />

</RelativeLayout>
```

toolbar.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.appcompat.widget.Toolbar
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="#FF9900"
    >

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:fontFamily="cursive"
        android:text="Practical 5"
        android:textSize="29dp"
        android:textStyle="bold" />

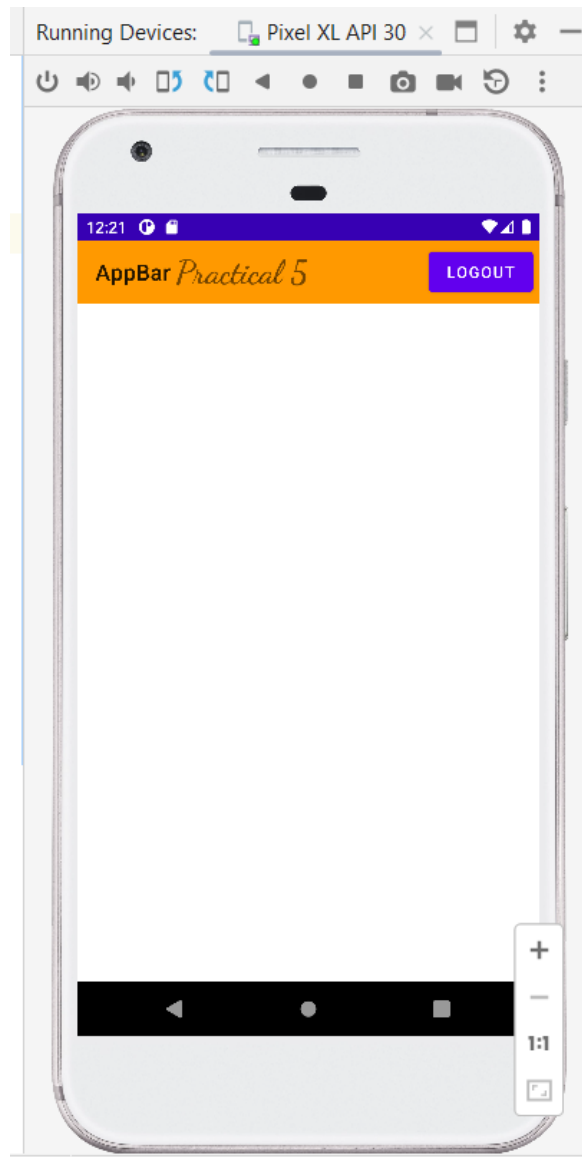
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginLeft="120dp"
        android:text="Logout" />
</androidx.appcompat.widget.Toolbar>
```

Advanced Mobile Programming

Main_Activity.java

```
package com.aayush.appbar;  
  
import android.os.Bundle;  
  
import androidx.appcompat.widget.Toolbar;  
import androidx.appcompat.app.AppCompatActivity;  
  
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
  
        Toolbar tbar = findViewById(R.id.t);  
        setSupportActionBar(tbar);  
    }  
}
```

Output:



5b. Demonstration of UI Components

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:layout_gravity="center"

    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="fill_parent"
        android:layout_height="fill_parent"
        android:textAlignment="center"
        android:orientation="vertical">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            android:text="Practical 5 : UI Components"
            android:textSize="30dp" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Calculator"
            android:layout_gravity="center"
            android:textSize="30dp" />

        <EditText
            android:id="@+id/et1"
            android:layout_width="match_parent"
            android:layout_height="50dp"
            android:hint="Number 1"
            android:inputType="number" />

        <EditText
            android:id="@+id/et2"
            android:layout_width="match_parent"
            android:layout_height="50dp"
            android:hint="Number 2"
            android:inputType="number"
            />

        <Button
            android:id="@+id/btnAdd"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Addition" />

        <Button
            android:id="@+id/btnSub"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Subtraction" />

    </LinearLayout>

</LinearLayout>
```

```
<Button
    android:id="@+id/btnMult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Multiplication" />

<Button
    android:id="@+id/btnDiv"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Division" />

<Button
    android:id="@+id/btnClear"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Clear" />

<TextView
    android:id="@+id/tv1"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Output :"
    android:textAlignment="center"
    android:textSize="30sp"
    android:textStyle="bold"
    />
</LinearLayout>
</LinearLayout>
```

MainActivity.java

```
package com.Aayush.a5_uicomponents;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText t1, t2;
    Button b1, b2, b3, b4, b5;
    TextView res;
    int n1 = 0, n2 = 0;
    String s1, s2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        t1 = (EditText) findViewById(R.id.et1);
        t2 = (EditText) findViewById(R.id.et2);

        b1 = (Button) findViewById(R.id.btnAdd);
        b2 = (Button) findViewById(R.id.btnSub);
        b3 = (Button) findViewById(R.id.btnMult);
        b4 = (Button) findViewById(R.id.btnDiv);
        b5 = (Button) findViewById(R.id.btnClear);

        res = (TextView) findViewById(R.id.tv1);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                try {
                    String s1 = t1.getText().toString();
                    String s2 = t2.getText().toString();
                    n1 = Integer.parseInt(s1);
                    n2 = Integer.parseInt(s2);
                    int sum = n1 + n2;
                    res.setText("Addition=" + sum);
                } catch (NumberFormatException e) {
                }
            }
        });

        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                try {
                    String s1 = t1.getText().toString();
                    String s2 = t2.getText().toString();
                    n1 = Integer.parseInt(s1);
                    n2 = Integer.parseInt(s2);
                    int sum = n1 - n2;
                    res.setText("Subtraction =" + sum);
                } catch (NumberFormatException e) {
                }
            }
        });
    }
}
```



```
        }
    }
});

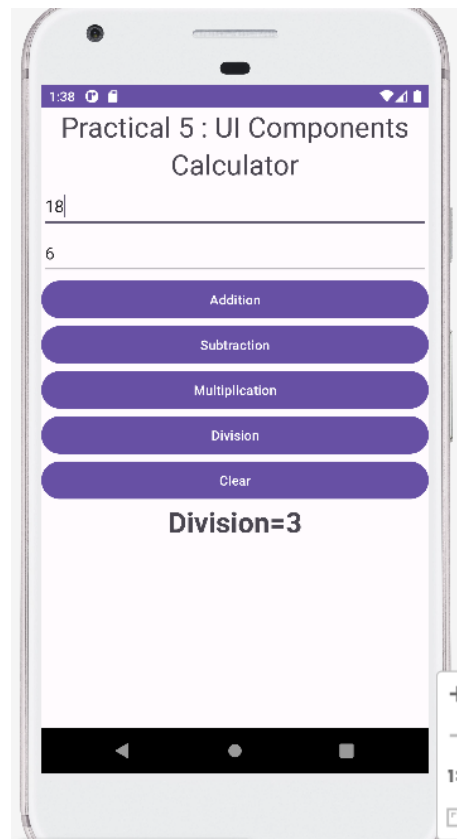
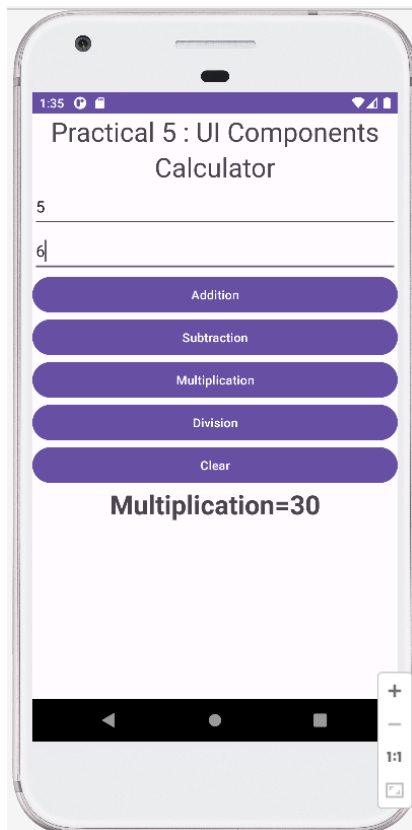
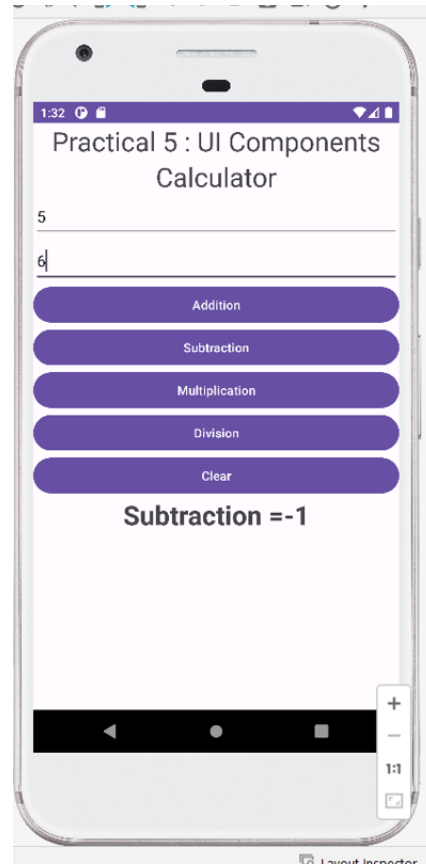
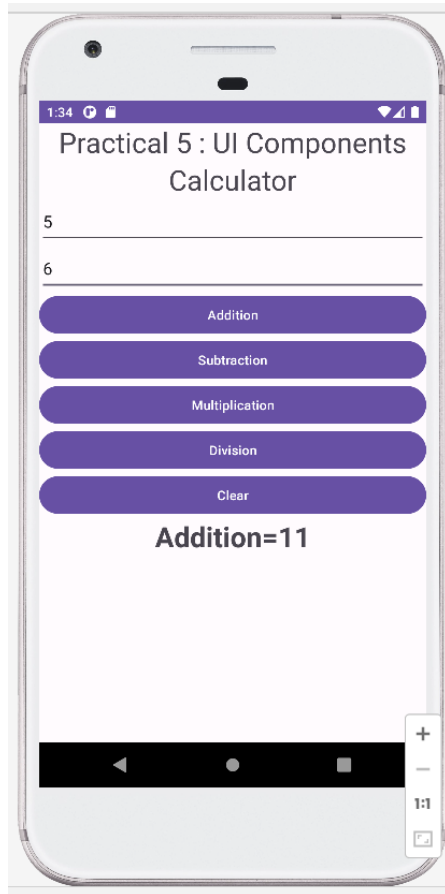
b3.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        try {
            String s1 = t1.getText().toString();
            String s2 = t2.getText().toString();
            n1 = Integer.parseInt(s1);
            n2 = Integer.parseInt(s2);
            int sum = n1 * n2;
            res.setText("Multiplication=" + sum);
        } catch (NumberFormatException e) {
        }
    }
});

b4.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        try {
            String s1 = t1.getText().toString();
            String s2 = t2.getText().toString();
            n1 = Integer.parseInt(s1);
            n2 = Integer.parseInt(s2);
            int sum = n1 / n2;
            res.setText("Division=" + sum);
        } catch (NumberFormatException e) {
        }
    }
});

b5.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        res.setText("Output");
    }
});
}
}
```

Advanced Mobile Programming

Output :



5c. Fragments

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        >

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/button1"
        android:text="News"
        />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/button2"
        android:text="Sports"
        />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/button3"
        android:text="Entertainment"
        />
</LinearLayout>
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <androidx.fragment.app.FragmentContainerView
        android:id="@+id/fragmentContainerView"
        android:name="com.aayush.a5b_fragments.News"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" >
    </androidx.fragment.app.FragmentContainerView>
</LinearLayout>
</LinearLayout>
```

Fragment_ent.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".EntFragment">

    <!-- TODO: Update blank fragment layout -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:textSize="50sp"
        android:textAlignment="center"
        android:text="Entertainment" />

</FrameLayout>
```

Fragment_news.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".News">

    <!-- TODO: Update blank fragment layout -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:textSize="50sp"
        android:textAlignment="center"
        android:text="News" />

</FrameLayout>
```

Fragment_sports.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".SportsFragment">

    <!-- TODO: Update blank fragment layout -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:textSize="50sp"
        android:textAlignment="center"
        android:text="Sports" />

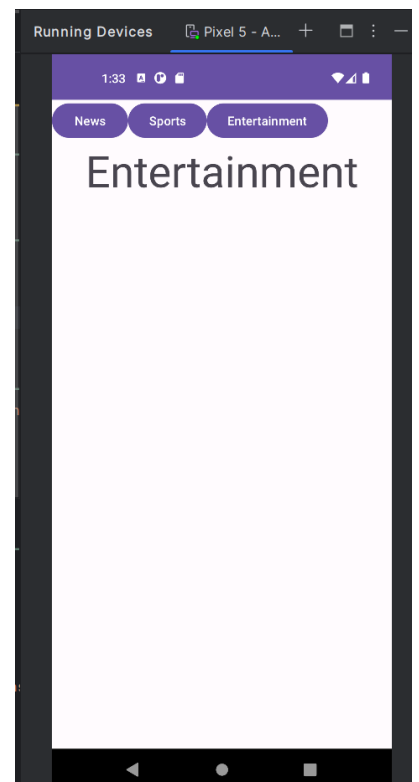
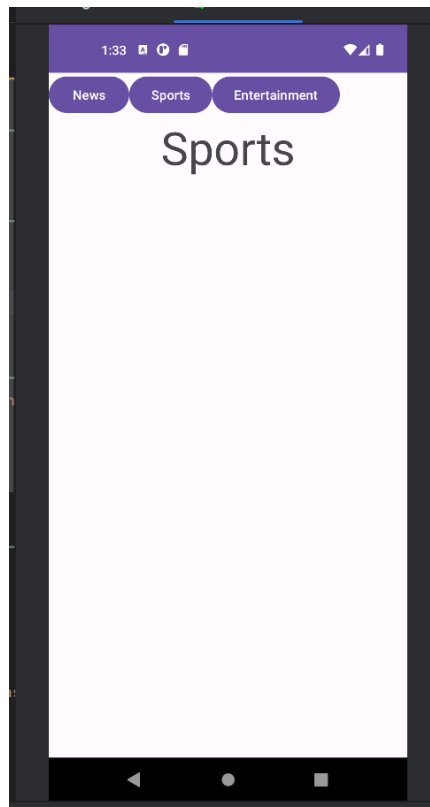
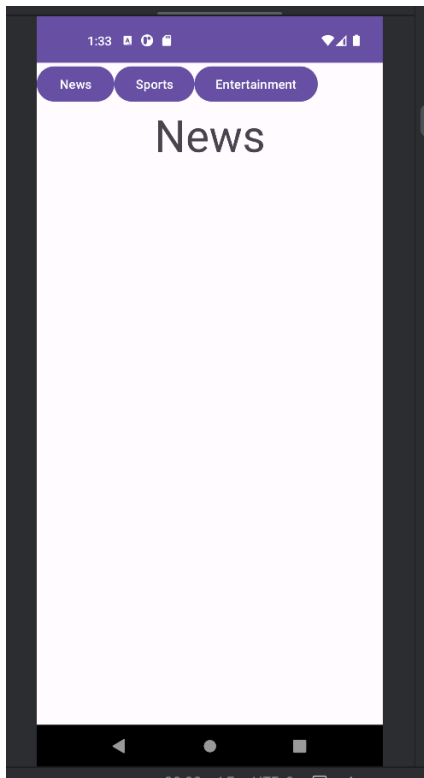
</FrameLayout>
```

MainActivity.java

```
package com.aayush.a5b_fragments;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentManager;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnnews=findViewById(R.id.button1);
        btnnews.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                FragmentManager fragmentManager = getSupportFragmentManager();
                fragmentManager.beginTransaction()
                    .replace(R.id.fragmentContainerView, News.class, null)
                    .setReorderingAllowed(true)
                    .addToBackStack("name") // Name can be null
                    .commit();
            }
        });
        Button btnsports=findViewById(R.id.button2);
        btnsports.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                FragmentManager fragmentManager = getSupportFragmentManager();
                fragmentManager.beginTransaction()
                    .replace(R.id.fragmentContainerView, SportsFragment.class, null)
                    .setReorderingAllowed(true)
                    .addToBackStack("name") // Name can be null
                    .commit();
            }
        });
        Button btnent=findViewById(R.id.button3);
        btnent.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                FragmentManager fragmentManager = getSupportFragmentManager();
                fragmentManager.beginTransaction()
                    .replace(R.id.fragmentContainerView, EntFragment.class, null)
                    .setReorderingAllowed(true)
                    .addToBackStack("name") // Name can be null
                    .commit();
            }
        });
    }
}
```

Output:



6. Menus

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

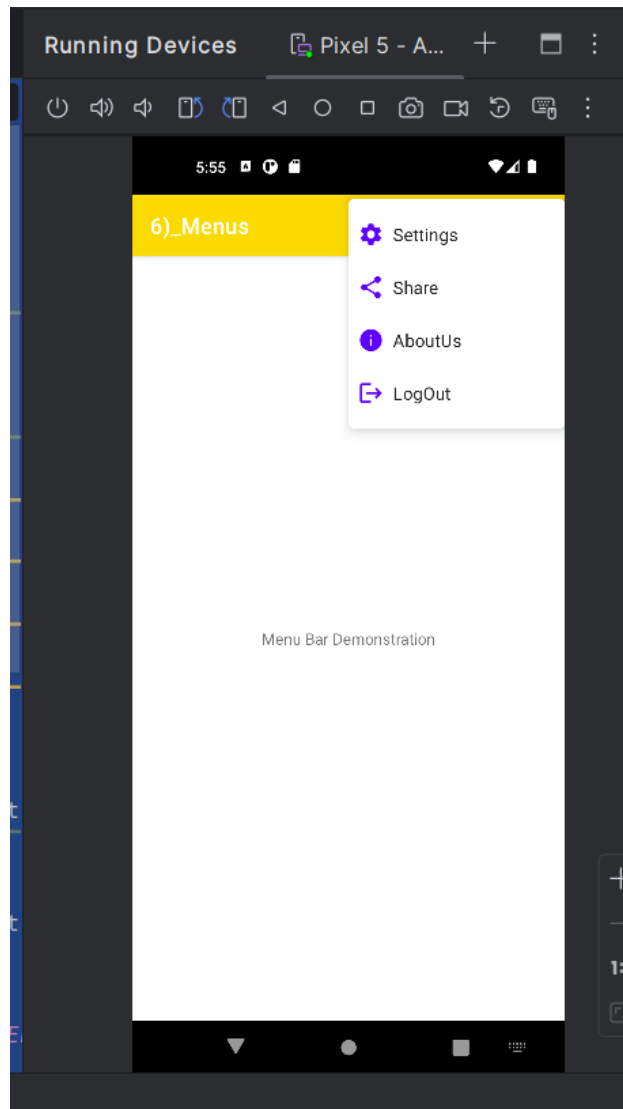
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu Bar Demonstration"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Main_Activity.java :

```
package com.aayush.a6_menus;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu,menu);
        return true;
    }
    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        switch (item.getItemId()){
            case R.id.settings:
                Toast.makeText(this,"You have clicked on settings",Toast.LENGTH_SHORT).show();
                return true;
            case R.id.aboutus:
                Toast.makeText(this,"You have clicked on About Us",Toast.LENGTH_SHORT).show();
                return true;
            case R.id.share:
                Toast.makeText(this,"You have clicked on Share",Toast.LENGTH_SHORT).show();
                return true;
            case R.id.logout:
                Toast.makeText(this,"You have clicked on Log Out",Toast.LENGTH_SHORT).show();
                return true;
            default:
                return super.onOptionsItemSelected(item);
        }
    }
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output :



7) Alert Dialog

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    tools:context=".MainActivity">
    <Button
        android:id="@+id/button"
        android:layout_width="141dp"
        android:layout_height="79dp"
        android:text="Click ME" />
</LinearLayout>
```

Main_Activity.java :

```
package com.aayush.a7_alertdialog;

import android.content.DialogInterface;
import android.content.Intent;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.view.View;
import android.widget.Toast;
import android.os.Bundle;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btn= findViewById(R.id.button);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                openAlertBox();
            }
        });
    }
    private void openAlertBox() {
        AlertDialog.Builder builder = new
            AlertDialog.Builder(MainActivity.this);
        builder.setMessage("Do you want to exit ?");
        builder.setIcon(R.drawable.baseline_crisis_alert_24);
        builder.setTitle("Alert !");

        builder.setCancelable(false);

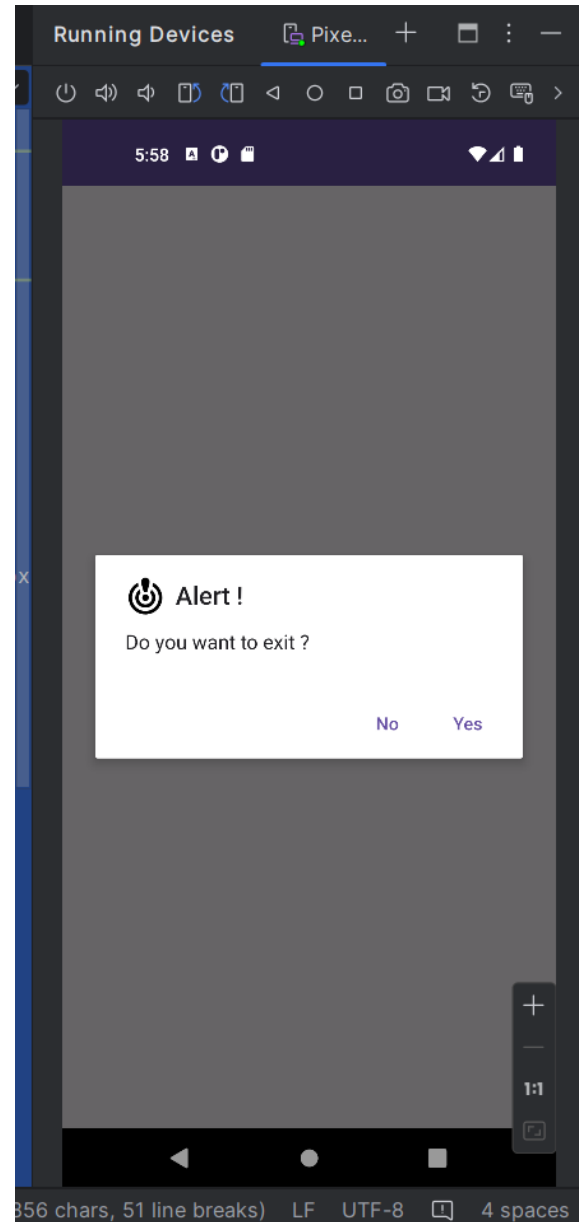
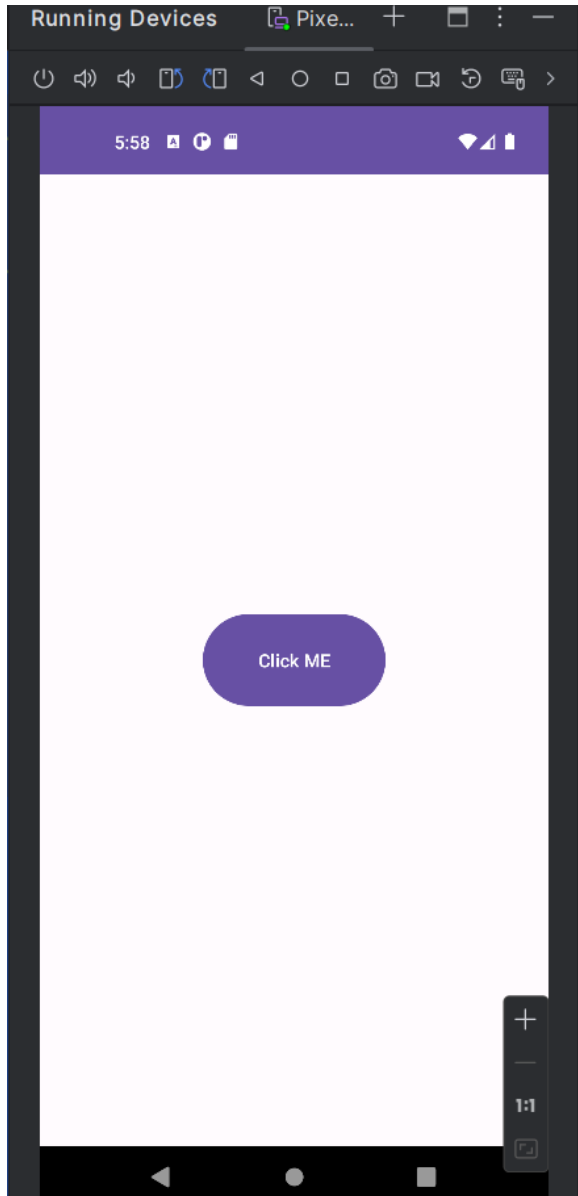
        builder.setPositiveButton("Yes", (DialogInterface.OnClickListener)
            (dialog, which) -> {
                // When the user click yes button then app will close
                finish();
            });

        builder.setNegativeButton("No", (DialogInterface.OnClickListener)
            (dialog, which) -> {
                // If user click no then dialog box is canceled.
                dialog.cancel();
            });
    }
}
```

Advanced Mobile Programming

```
// Create the Alert dialog
AlertDialog alertDialog = builder.create();
// Show the Alert Dialog box
alertDialog.show();
    }
}
```

Output :



Practical 8

Database Programming with SQLite

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentStart="true"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:text="Name :"
        android:textAppearance="?android:attr/textAppearanceLarge" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/editText_name"
        android:layout_alignParentStart="true"
        android:layout_alignParentLeft="true"
        android:text="Surname :"
        android:textAppearance="?android:attr/textAppearanceLarge" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/editText_surname"
        android:layout_alignParentStart="true"
        android:layout_alignParentLeft="true"
        android:text="Marks :"
        android:textAppearance="?android:attr/textAppearanceLarge" />

    <EditText
        android:id="@+id/editText_name"
        android:layout_width="match_parent"
        android:layout_height="50dp"
        android:layout_alignTop="@+id/textView"
        android:layout_toEndOf="@+id/textView"
        android:layout_toRightOf="@+id/textView"
        android:hint="Enter Your Name" />

    <EditText
        android:id="@+id/editText_surname"
        android:layout_width="match_parent"
        android:layout_height="50dp"
        android:layout_alignTop="@+id/textView2"
        android:layout_toEndOf="@+id/textView2"
        android:layout_toRightOf="@+id/textView2"
        android:hint="Enter Your Surname" />

    <EditText
        android:id="@+id/editText_Marks"
        android:layout_width="match_parent"
        android:layout_height="50dp"
```

```
android:layout_below="@+id/editText_surname"
android:layout_toEndOf="@+id/textView3"
android:layout_toRightOf="@+id/textView3"
android:hint="Enter The Marks" />
```

```
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add Data"
    android:id="@+id/button_add"
    android:layout_below="@+id/editText_Marks"
    android:layout_alignParentLeft="true"
    android:layout_alignParentStart="true"
    android:layout_marginTop="76dp" />
```

```
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="View All"
    android:id="@+id/button_viewAll"
    android:layout_above="@+id/button_update"
    android:layout_centerHorizontal="true" />
```

```
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Update"
    android:id="@+id/button_update"
    android:layout_below="@+id/button_add"
    android:layout_alignParentLeft="true"
    android:layout_alignParentStart="true" />
```

```
<TextView
    android:id="@+id/textView_id"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

    android:layout_below="@+id/editText_Marks"
    android:layout_alignParentStart="true"
    android:layout_alignParentLeft="true"
    android:text="id :"
    android:textAppearance="?android:attr/textAppearanceLarge" />
```

```
<EditText
    android:id="@+id/editText_id"
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_alignTop="@+id/textView_id"
    android:layout_toEndOf="@+id/textView3"
    android:layout_toRightOf="@+id/textView3"
    android:hint="Enter ID to Update Or Delete !" />
```

```
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Delete"
    android:id="@+id/button_delete"
    android:layout_centerVertical="true"
    android:layout_below="@+id/button_viewAll"
    android:layout_alignLeft="@+id/button_viewAll"
    android:layout_alignStart="@+id/button_viewAll" />
```

```
</RelativeLayout>
```

MainActivity.java

```
package com.example.a8_database;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    DatabaseHelper myDb;
    EditText editName,editSurname,editMarks ,editTextId;
    Button btnAddData;
    Button btnviewAll;
    Button btnDelete;
    Button btnviewUpdate;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        myDb = new DatabaseHelper(this);
        editName = (EditText)findViewById(R.id.editText_name);
        editSurname = (EditText)findViewById(R.id.editText_surname);
        editMarks = (EditText)findViewById(R.id.editText_Marks);
        editTextId = (EditText)findViewById(R.id.editText_id);
        btnAddData = (Button)findViewById(R.id.button_add);
        btnviewAll = (Button)findViewById(R.id.button_viewAll);
        btnviewUpdate= (Button)findViewById(R.id.button_update);
        btnDelete= (Button)findViewById(R.id.button_delete);
        AddData();
        viewAll();
        UpdateData();
        DeleteData();
    }
    public void DeleteData() {
        btnDelete.setOnClickListener(
            new View.OnClickListener() {
                @Override
                public void onClick(View v) {
                    Integer deletedRows = myDb.deleteData(editTextId.getText().toString());
                    if(deletedRows > 0)
                        Toast.makeText(MainActivity.this,"Data Deleted",Toast.LENGTH_LONG).show();
                    else
                        Toast.makeText(MainActivity.this,"ID Not Found",Toast.LENGTH_LONG).show();
                }
            }
        );
    }
    public void UpdateData() {
        btnviewUpdate.setOnClickListener(
            new View.OnClickListener() {
                @Override
                public void onClick(View v) {
                    boolean isUpdate = myDb.updateData(editTextId.getText().toString(),
                        editName.getText().toString(),
                        editSurname.getText().toString(),editMarks.getText().toString());
                    if(isUpdate == true)
                        Toast.makeText(MainActivity.this,"Data Updated",Toast.LENGTH_LONG).show();
                }
            }
        );
    }
}
```

Advanced Mobile Programming

```
        else
            Toast.makeText(MainActivity.this,"Data not Updated",Toast.LENGTH_LONG).show();
    }
}

);

}

public void AddData() {
    btnAddData.setOnClickListener(
        new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                boolean isInserted = myDb.insertData(editName.getText().toString(),
                    editSurname.getText().toString(),
                    editMarks.getText().toString() );
                if(isInserted == true)
                    Toast.makeText(MainActivity.this,"Data Inserted",Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(MainActivity.this,"Data not Inserted",Toast.LENGTH_LONG).show();
            }
        }
    );
}

public void viewAll() {
    btnviewAll.setOnClickListener(
        new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Cursor res = myDb.getAllData();
                if(res.getCount() == 0) {
                    // show message
                    showMessage("Error","Nothing found");
                    return;
                }
                StringBuffer buffer = new StringBuffer();
                while (res.moveToNext()) {
                    buffer.append("Id :"+ res.getString(0)+"\n");
                    buffer.append("Name :"+ res.getString(1)+"\n");
                    buffer.append("Surname :"+ res.getString(2)+"\n");
                    buffer.append("Marks :"+ res.getString(3)+"\n\n");
                }
                // Show all data
                showMessage("Data",buffer.toString());
            }
        }
    );
}

}

public void showMessage(String title,String Message){
    AlertDialog.Builder builder = new AlertDialog.Builder(this);
    builder.setCancelable(true);
    builder.setTitle(title);
    builder.setMessage(Message);
    builder.show();
}

}
```

DatabaseHelper.java

```
package com.example.a8_database;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

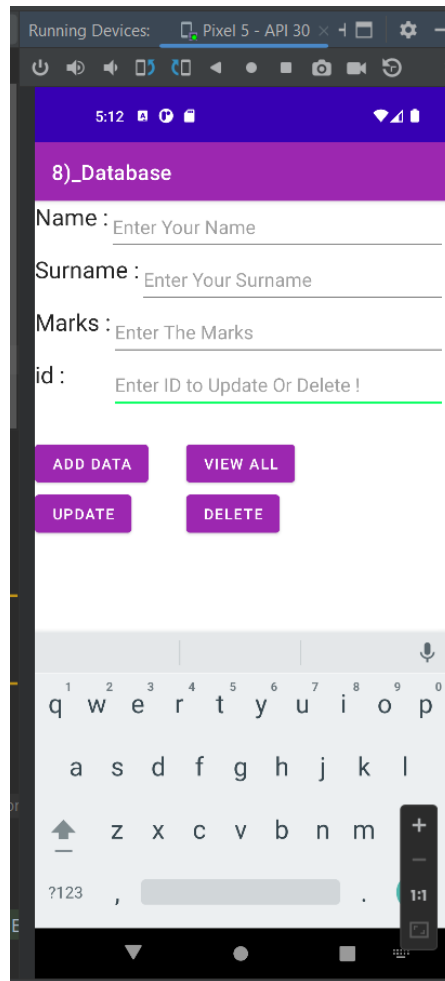
public class DatabaseHelper extends SQLiteOpenHelper {
    public static final String DATABASE_NAME = "Student.db";
    public static final String TABLE_NAME = "student_table";
    public static final String COL_1 = "ID";
    public static final String COL_2 = "NAME";
    public static final String COL_3 = "SURNAME";
    public static final String COL_4 = "MARKS";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, 1);
    }
    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("create table " + TABLE_NAME + " (ID INTEGER PRIMARY KEY AUTOINCREMENT,NAME TEXT,SURNAME TEXT,MARKS INTEGER)");
    }

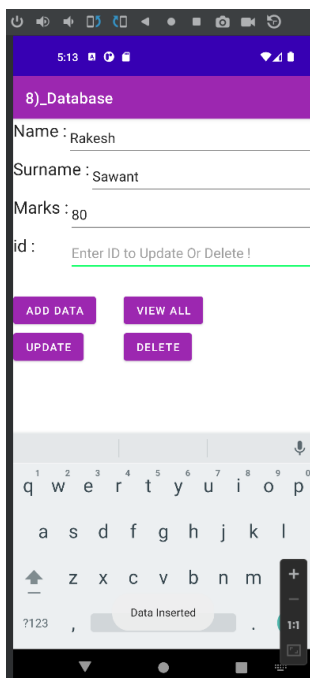
    @Override
    public void onUpgrade(SQLiteDatabase db, int i, int i1) {
        db.execSQL("DROP TABLE IF EXISTS "+TABLE_NAME);
        onCreate(db);
    }
    public boolean insertData(String name,String surname,String marks) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COL_2,name);
        contentValues.put(COL_3,surname);
        contentValues.put(COL_4,marks);
        long result = db.insert(TABLE_NAME,null ,contentValues);
        if(result == -1)
            return false;
        else
            return true;
    }
    public Cursor getAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        Cursor res = db.rawQuery("select * from "+TABLE_NAME,null);
        return res;
    }
    public boolean updateData(String id,String name,String surname,String marks) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COL_1,id);
        contentValues.put(COL_2,name);
        contentValues.put(COL_3,surname);
        contentValues.put(COL_4,marks);
        db.update(TABLE_NAME, contentValues, "ID = ?",new String[] { id });
        return true;
    }
    public Integer deleteData (String id) {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.delete(TABLE_NAME, "ID = ?",new String[] {id});
    }
}
```

Advanced Mobile Programming

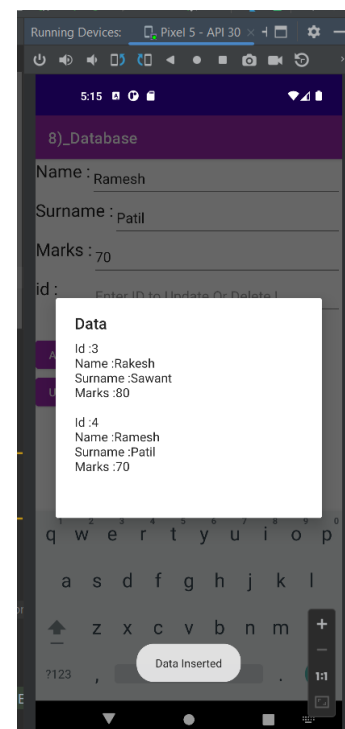
Output :



Insert Data

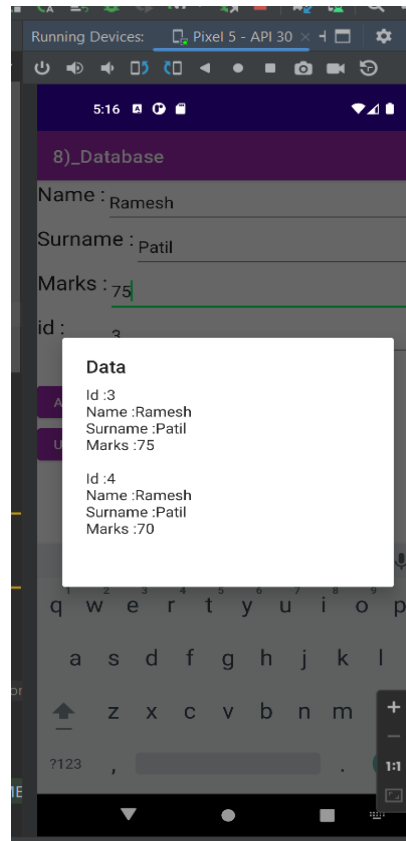
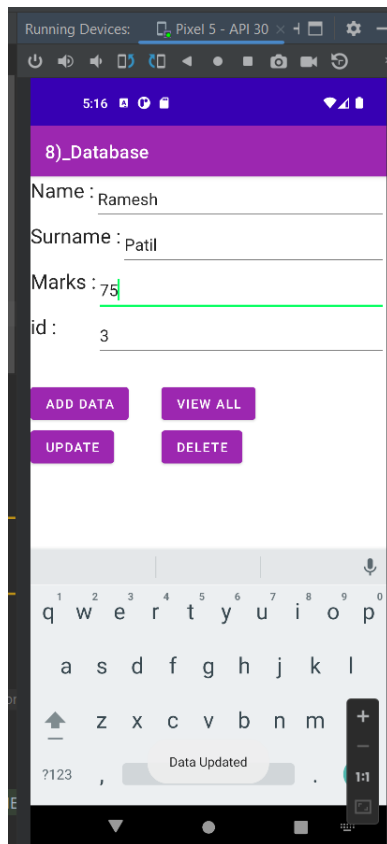


View All



Advanced Mobile Programming

Update Data :



Delete Data :

